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## HTTP communication mode

### Set & Get Quota

How to transfer request parameters by these two HTTP API.

PUT: /iot-open/sign/device/quota: SetCmdRequest

GET: /iot-open/sign/device/quota: Get cmdRequest, GetCmdResponse

| Set Command | SetCmdRequest | GetCmdRequest | GetCmdResponse |
|---|---|---|---|
| Setting the X-Boost switch | `"params":{ "cmdSet": 32, "id": 66, "enabled": 0, "xboost": 0 }` | `"params":{ "cmdSet": 32, "id": 66, "quotas": [ "inv.cfgAcEnabled", "inv.cfgAcXboost" ] }` | `"data":{ "inv.cfgAcEnabled": 0, "inv.cfgAcXboost": 0 }` |
| Setting the car charger switch | `"params":{ "cmdSet": 32, "id": 81, "enabled": 1 }` | `"params":{ "cmdSet": 32, "id": 81, "quotas": [ "mppt.carState" ] }` | `"data":{ "mppt.carState": 1 }` |
| Setting the charge level | `"params":{ "cmdSet": 32, "id": 49, "maxChgSoc": 100 }` | `"params":{ "cmdSet": 32, "id": 49, "quotas": [ "ems.maxChargeSoc" ] }` | `"data":{ "ems.maxChargeSoc": 0 }` |

| Set Command | SetCmdRequest | GetCmdRequest | GetCmdResponse |
| --- | --- | --- | --- |
| Setting the discharge level | `"params":{ "cmdSet": 32, "id": 51, "minDsgSoc": 10 }` | `"params":{ "cmdSet": 32, "id": 51, "quotas": [ "ems.minDsgSoc" ] }` | `"data":{ "ems.minDsgSoc": 0 }` |
| Setting the car input current | `"params":{ "cmdSet": 32, "id": 71, "currMa": 4000 }` | `"params":{ "cmdSet": 32, "id": 71, "quotas": [ "mppt.cfgDcChgCurrent" ] }` | `"data":{ "mppt.cfgDcChgCurrent": 0 }` |
| Setting the beep switch | `"params":{ "cmdSet": 32, "id": 38, "enabled": 1 }` | `"params":{ "cmdSet": 32, "id": 38, "quotas": [ "pd.beepState" ] }` | `"data":{ "pd.beepState": 1 }` |
| Setting the screen brightness | `"params":{ "cmdSet": 32, "id": 39, "lcdBrightness": 100 }` | `"params":{ "cmdSet": 32, "id": 39, "quotas": [ "pd.lcdBrightness" ] }` | `"data":{ "pd.lcdBrightness": 100 }` |

| Set Command | SetCmdRequest | GetCmdRequest | GetCmdResponse |
| --- | --- | --- | --- |
| Setting the lower threshold percentage of smart generator auto on | `"params":{ "cmdSet": 32, "id": 52, "openOilSoc": 52 }` | `"params":{ "cmdSet": 32, "id": 52, "quotas": [ "ems.minOpenOilEbSoc" ] }` | `"data":{ "ems.minOpenOilEbSoc": 0 }` |
| Setting the upper threshold percentage of smart generator auto off | `"params":{ "cmdSet": 32, "id": 53, "closeOilSoc": 10 }` | `"params":{ "cmdSet": 32, "id": 53, "quotas": [ "ems.maxCloseOilEbSoc" ] }` | `"data":{ "ems.maxCloseOilEbSoc": 0 }` |
| Setting the unit timeout | `"params":{ "cmdSet": 32, "id": 33, "standByMode": 0 }` | `"params":{ "cmdSet": 32, "id": 33, "quotas": [ "pd.standByMode" ] }` | `"data":{ "pd.standByMode": 0 }` |
| Setting the screen timeout | `"params":{ "cmdSet": 32, "id": 39, "lcdTime": 60 }` | `"params":{ "cmdSet": 32, "id": 39, "quotas": [ "pd.lcdOffSec" ] }` | `"data":{ "pd.lcdOffSec": 60 }` |

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| Set Command | SetCmdRequest | GetCmdRequest | GetCmdResponse |
| --- | --- | --- | --- |
| Setting the AC standby time | "params":{<br>"cmdSet": 32,<br>"id": 153,<br>"standByMins": 720<br>} | "params":{<br>"cmdSet": 32,<br>"id": 153,<br>"quotas": [<br>"inv.cfgStandbyMin"<br>]<br>} | "data":{<br>"inv.cfgStandbyMin": 720<br>} |
| AC charging settings | "params":{<br>"cmdSet": 32,<br>"id": 69,<br>"slowChgPower": 0<br>} | "params":{<br>"cmdSet": 32,<br>"id": 69,<br>"quotas": [<br>"inv.cfgSlowChgWatts",<br>"inv.acDipSwitch"<br>]<br>} | "data":{<br>"inv.cfgSlowChgWatts": 0,<br>"inv.acDipSwitch": 0<br>} |
| PV charging type | "params":{<br>"cmdSet": 32,<br>"id": 82,<br>"chgType": 0<br>} | "params":{<br>"cmdSet": 32,<br>"id": 82,<br>"quotas": [<br>"mppt.cfgChgType"<br>]<br>} | "data":{<br>"mppt.cfgChgType": 0<br>} |
| Bypass AC auto start | "params":{<br>"cmdSet": 32,<br>"id": 84,<br>"enabled": 0<br>} | "params":{<br>"cmdSet": 32,<br>"id": 84,<br>"quotas": [<br>"inv.acPassbyAutoEn"<br>]<br>} | "data":{<br>"inv.acPassbyAutoEn": 0<br>} |

Example

PUT: /iot-open/sign/device/quota: SetCmdRequest

```
// Delta Pro: Setting the X-Boost switch
// Request
curl -X PUT https://api-e.ecoflow.com/iot-open/sign/device/quota \
-H 'Content-Type:application/json;charset=UTF-8' \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:0f3a1b102cd9a306307b7a9a0f0a9add7b8bfc93cf11***' \
-d '{"sn":"DCABZ***","params":{"cmdSet":32,"id":66,"enabled":1}}'

// Response
{
    "code":"0",
    "message":"Success"
}
```

GET: /iot-open/sign/device/quota: Get cmdRequest, GetCmdResponse

```
// Delta Pro: Get the X-Boost switch value
// Request
curl -X POST https://api-e.ecoflow.com/iot-open/sign/device/quota \
-H 'Content-Type:application/json;charset=UTF-8' \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:1e48307441d06a6b31d4c0d66be1580c7748fd0***' \
-d '{"sn":"DCABZ***","params":{"cmdSet":32,"id":66,"quotas":["inv.cfgAcEnabled"]}}'

// Response
{
    "code":"0",
    "message":"Success",
    "data":{
        "inv.cfgAcEnabled":1
    }
}
```

**GetAllQuotaResponse**

How to transfer request parameters by this HTTP API.

GET: /iot-open/sign/device/quota/all: GetAllQuotaResponse

| Field | Field's Type | Description |
| --- | --- | --- |
| bmsMaster.soc | short | Remaining battery percentage |
| bmsMaster.temp | byte | Temperature ℃ |
| bmsMaster.inputWatts | long | Input power |
| bmsMaster.outputWatts | long | Output power |
| bmsMaster.num | short | BMS number |
| bmsMaster.type | short | BMS type: 1. Lithium battery; 2. Oil-powered |
| bmsMaster.cellId | short | Battery capacity type: 1: 2.5 Ah per battery; 2: 2 Ah per battery |
| bmsMaster.errCode | long | Global error code |
| bmsMaster.sysVer | long | System version |

| Field | Field's Type | Description |
| --- | --- | --- |
| bmsMaster.vol | float | Voltage |
| bmsMaster.amp | float | Current |
| bmsMaster.openBmsIdx | short | Battery pack enable state |
| bmsMaster.designCap | long | Design capacity (mAh) |
| bmsMaster.remainCap | long | Remaining capacity (mAh) |
| bmsMaster.fullCap | long | Full capacity (mAh) |
| bmsMaster.soh | short | Health status |
| bmsMaster.maxCellVol | int | Maximum cell voltage |
| bmsMaster.minCellVol | int | Minimum cell voltage |
| bmsMaster.maxCellTemp | byte | Maximum cell temperature |
| bmsMaster.minCellTemp | byte | Minimum cell temperature |
| bmsMaster.maxMosTemp | byte | Maximum MOS temperature |
| bmsMaster.minMosTemp | byte | Minimum MOS temperature |
| bmsMaster.bmsFault | short | BMS permanent fault |
| bmsMaster.bqSysStatReg | short | BQ hardware protection register |
| bmsMaster.tagChgAmp | long | Target charging current |
| bmsMaster.f32ShowSoc | float | SOC |
| bmsMaster.remainTime | long | Time remaining |
| inv.errCode | long | Global error code |
| inv.sysVer | long | System version |
| inv.chargerType | short | Charger type: 1: AC charging; 2: DC adapter charging; 3: solar charging; 4: CC; 5: BC |
| inv.inputWatts | int | Charging power (W) |
| inv.outputWatts | int | Discharging power (W) |
| inv.invType | short | PSDR model code |
| inv.invOutVol | long | Actual inverter output voltage (mV) |
| inv.invOutAmp | long | Inverter output current (mA) |

| Field | Field's Type | Description |
| --- | --- | --- |
| inv.invOutFreq | short | Inverter output frequency (Hz) |
| inv.acInVol | long | Inverter input voltage (mV) |
| inv.acInAmp | long | Inverter input current (mA) |
| inv.acInFreq | short | Inverter input frequency (Hz) |
| inv.outTemp | short | Inverter temperature (℃) |
| inv.dcInVol | long | DC input voltage (mV) |
| inv.dcInAmp | long | DC input current (mA) |
| inv.dcInTemp | short | DC temperature (℃) |
| inv.fanState | short | Fan state: 0: disabled; 1: Level 1; 2: Level 2; 3: Level 3 |
| inv.cfgAcEnabled | short | AC discharge switch setting 0 off, 1 on |
| inv.cfgAcXboost | short | X-Boost switch: 0: off; 1: on; 0xff: ignored |
| inv.cfgAcOutVoltage | long | Inverter output voltage (V): 0xffffffff: ignored |
| inv.cfgAcOutFreq | short | Inverter output frequency (Hz): 1: 50 Hz; 2: 60 Hz; 0xff: ignored |
| inv.cfgAcWorkMode | short | AC charging mode: 0: full power; 1: mute |
| inv.chgPauseFlag | short | AC charging pause flag: 1: charging stopped |
| inv.acDipSwitch | short | AC fast/slow charging dip switch: 0: unknown; 1: fast charging mode; 2: slow charging mode |
| inv.cfgFastChgWatts | int | Maximum charging power for AC fast charging (W): DELTA2000 (100 W–1400 W); DELTA MINI (TBD); DELTA3000 (TBD) |
| inv.cfgSlowChgWatts | int | Maximum charging power for AC slow charging (W): DELTA2000 (100 W–700 W); DELTA MINI (TBD); DELTA3000 (TBD) |
| inv.cfgStandbyMin | int | *AC standby time /min*** |

| Field | Field's Type | Description |
| --- | --- | --- |
| | | *0 Never standby*** *720 Default value* |
| inv.dischargeType | short | Discharging type: 1: AC discharging; 2: PR; 3: BC |
| mppt.faultCode | long | Error code: byte0: mppt_fault; byte1: car_fault; byte2: dc24v_fault |
| mppt.swVer | long | Version number |
| mppt.inVol | long | PV input voltage (mV) |
| mppt.inAmp | long | PV input current (mA) |
| mppt.inWatts | int | PV input power (W) |
| mppt.outVol | long | PV output voltage (mV) |
| mppt.outAmp | long | PV output current (mA) |
| mppt.outWatts | int | PV output power (W) |
| mppt.mpptTemp | short | MPPT temperature (℃) |
| mppt.xt60ChgType | short | XT60 charging type: 0: not detected; 1: MPPT; 2: adapter |
| mppt.cfgChgType | short | Configured charging type: This parameter is valid when xt60_chg_type is 0. 0: auto; 1: MPPT; 2: adapter |
| mppt.chgType | short | Actual charging type: 0: null; 1: adapter (adapter/DC source); 2: MPPT (solar); 3: AC (mains supply); 4: gas; 5: wind |
| mppt.chgState | short | Charging state: 0: disabled; 1: charging; 2: standby (DC charging stopped during AC charging) |
| mppt.dcdc12vVol | long | DC12V30A output voltage (mV), which is valid only for DELTA Pro |
| mppt.dcdc12vAmp | long | DC12V30A output current (mA), which is valid only for DELTA Pro |
| mppt.dcdc12vWatts | int | DC12V30A output power (W), which is valid only for DELTA Pro |

| Field | Field's Type | Description |
| --- | --- | --- |
| mppt.carOutVol | long | Car charging output voltage (mV) |
| mppt.carOutAmp | long | Car charging output current (mA) |
| mppt.carOutWatts | int | Car charging output power (W) |
| mppt.carTemp | short | Car charging temperature (℃) |
| mppt.carState | short | Car charger switch settin<br>0 off, 1 on |
| mppt.dc24vTemp | short | DCDC24V temperature (℃) |
| mppt.dc24vState | short | DCDC24 switch state: 0: off; 1: on |
| mppt.chgPauseFlag | short | PV charging pause flag: 1: charging stopped |
| mppt.cfgDcChgCurrent | long | On-board charging current /mA |
| mppt.reserved | byte[] | Reserved |
| pd.model | short | Product model |
| pd.errCode | long | Global error code |
| pd.sysVer | long | System version |
| pd.wifiVer | long | Wi-Fi version |
| pd.wifiAutoRcvy | short | Wi-Fi auto mode: 0: default mode (STA); 1: The Wi-Fi network is automatically restored to the last mode (STA/AP) after powering on. |
| pd.soc | short | Displayed SOC |
| pd.wattsOutSum | int | Total output power (W) |
| pd.wattsInSum | int | Total input power (W) |
| pd.remainTime | int | Time remaining (min) > 0: remaining charging time; time remaining (min) < 0: remaining discharging time |
| pd.beepState | short | Beep status<br>0 Normal, 1 Quiet |

| Field | Field's Type | Description |
| --- | --- | --- |
| pd.dcOutState | short | DC button state: 0: off; 1: on |
| pd.usb1Watts | short | Common usb1 output power /W |
| pd.usb2Watts | short | Common usb2 output power /W |
| pd.qcUsb1Watts | short | Quick charge usb1 output power /W |
| pd.qcUsb2Watts | short | Quick charge usb2 output power /W |
| pd.typec1Watts | short | Typec1 output power /W |
| pd.typec2Watts | short | Typec2 output power /W |
| pd.typec1Temp | byte | Type-C 1 temperature (℃) |
| pd.typec2Temp | byte | Type-C 2 temperature (℃) |
| pd.carState | short | CAR button state: 0: off; 1: on |
| pd.carWatts | short | CAR output power (W) |
| pd.carTemp | byte | CAR temperature (℃) |
| pd.standByMode | int | Device standby time /min<br>0 Never standby<br>5999 Max value |
| pd.lcdOffSec | int | LCD screen-off duration: 0: never off |
| pd.brightnessLevel | short | LCD brightness level: 0–3 |
| pd.chgPowerDc | long | Cumulative DC power charged (Wh) (adapter) |
| pd.chgSunPower | long | Cumulative solar power charged (Wh) |
| pd.chgPowerAc | long | Cumulative AC power charged (Wh) (wall socket) |
| pd.dsgPowerDc | long | Cumulative DC power discharged (Wh) |
| pd.dsgPowerAc | long | Cumulative AC power discharged (Wh) |
| pd.usbUsedTime | long | USB use time (s) |
| pd.usbqcUsedTime | long | USB QC use time (s) |

| Field | Field's Type | Description |
| --- | --- | --- |
| pd.typccUsedTime | long | Type-C use time (s) |
| pd.carUsedTime | long | Car use time (s) |
| pd.invUsedTime | long | Inverter use time (s) |
| pd.dcInUsedTime | long | DC charging time (s) |
| pd.mpptUsedTime | long | MPPT use time (s) |
| pd.extRj45Port | short | RJ45 port |
| pd.ext3p8Port | short | Infinity port |
| pd.ext4p8Port | short | Extra battery port. Only the status of the leftmost port can be identified. |
| pd.sysChgDsgState | short | Charging/discharging state on screen: 1: discharged; 2: charged |
| pd.wifiRssi | byte | Wi-Fi signal intensity |
| pd.wirelessWatts | short | Wireless charging output power (W): Reserved, not available |
| pd.iconRechgTimeMode | byte | Charge icon mode: 0: normal; 1: blinking |
| pd.iconRechgTimeState | byte | Charge icon state: 0: off; 1: on. This parameter is valid only when the charge icon mode is 0. |
| pd.iconFanMode | byte | Fan icon mode: 0: normal; 1: blinking |
| pd.iconFanState | byte | Fan icon state: 0: off; 1: Level 1; 2: Level 2; 3: Level 3. This parameter is valid only when the fan icon mode is 0. |
| pd.iconBmsParallelMode | byte | BMS parallel icon mode: 0: normal; 1: blinking |
| pd.iconBmsParallelState | byte | BMS parallel icon state: 0: off; 1: on. This parameter is valid only when the BMS parallel icon mode is 0. |
| pd.iconInvParallelMode | byte | Inverter parallel icon mode: 0: normal; 1: blinking |
| pd.iconInvParallelState | byte | Inverter parallel icon state: 0: off; |

| Field | Field's Type | Description |
| --- | --- | --- |
| | | 1: on. This parameter is valid only when the inverter parallel icon mode is 0. |
| pd.iconAcFreqMode | byte | AC icon mode: 0: normal; 1: blinking |
| pd.iconAcFreqState | byte | AC icon state: 0: off; 1: 50 Hz; 2: 60 Hz. This parameter is valid only when the AC icon mode is 0. |
| pd.iconSocUpsMode | byte | UPS icon mode: 0: normal; 1: blinking |
| pd.iconSocUpsState | byte | UPS icon state: 0: off; 1: on. This parameter is valid only when the UPS icon mode is 0. |
| pd.iconUsbMode | byte | USB icon mode: 0: normal; 1: blinking |
| pd.iconUsbState | byte | USB icon state: 0: off; 1: on. This parameter is valid only when the USB icon mode is 0. |
| pd.iconTypecMode | byte | Type-C icon mode: 0: normal; 1: blinking |
| pd.iconTypecState | byte | Type-C icon state: 0: off; 1: on. This parameter is valid only when the Type-C icon mode is 0. |
| pd.iconCarMode | byte | CAR icon mode: 0: normal; 1: blinking |
| pd.iconCarState | byte | CAR icon state: 0: off; 1: on. This parameter is valid only when the CAR icon mode is 0. |
| pd.iconWifiMode | byte | Wi-Fi icon mode: 0: normal; 1: blinking |
| pd.iconWifiState | byte | Wi-Fi icon state: 0: off; 1: on. This parameter is valid only when the Wi-Fi icon mode is 0. |
| pd.iconBmsErrMode | byte | Exclamation mark icon mode: 0: normal; 1: blinking |
| pd.iconBmsErrState | byte | Exclamation mark icon state: 0: off; 1: on. This parameter is valid |

| Field | Field's Type | Description |
| --- | --- | --- |
| | | only when the exclamation mark icon mode is 0. |
| pd.iconOverloadMode | byte | OVERLOAD icon mode: 0: normal; 1: blinking |
| pd.iconOverloadState | byte | OVERLOAD icon state: 0: off; 1: on. This parameter is valid only when the OVERLOAD icon mode is 0. |
| pd.iconHiTempMode | byte | High temperature icon mode: 0: normal; 1: blinking |
| pd.iconHiTempState | byte | High temperature icon state: 0: off; 1: on. This parameter is valid only when the high temperature icon mode is 0. |
| pd.iconLowTempMode | byte | Low temperature icon mode: 0: normal; 1: blinking |
| pd.iconLowTempState | byte | Low temperature icon state: 0: off; 1: on. This parameter is valid only when the low temperature icon mode is 0. |
| pd.iconWirelessChgMode | byte | Wireless charging icon mode: 0: normal; 1: blinking |
| pd.iconWirelessChgState | byte | Wireless charging icon state: 0: off; 1: on. This parameter is valid only when the wireless charging icon mode is 0. |
| pd.iconPackHeaterMode | byte | Battery heater icon mode: 0: normal; 1: blinking |
| pd.iconPackHeaterState | byte | Battery heater icon state: 0: off; 1: on. This parameter is valid only when the battery heater icon mode is 0. |
| pd.iconFactoryMode | byte | Factory icon mode: 0: normal; 1: blinking |
| pd.iconFactoryState | byte | Factory icon state: 0: off; 1: on. This parameter is valid only when the factory icon mode is 0. |
| pd.iconBtMode | byte | Bluetooth icon mode: 0: normal; |

| Field | Field's Type | Description |
| --- | --- | --- |
| | | 1: blinking |
| pd.iconBtState | byte | Bluetooth icon state: 0: off; 1: on. This parameter is valid only when the Bluetooth icon mode is 0. |
| pd.iconRcMode | byte | Remote control icon mode: 0: normal; 1: blinking |
| pd.iconRcState | byte | Remote control icon state: 0: off; 1: on; 2: one signal bar; 3: two signal bars; 4: searching signal. This parameter is valid only when the remote control icon mode is 0. |
| pd.iconChgStationMode | byte | Charging pile icon mode: 0: normal; 1: blinking |
| pd.iconChgStationState | byte | Charging pile icon state: 0: off; 1: on. This parameter is valid only when the charging pile icon mode is 0. |
| pd.iconCoGasMode | byte | CO toxic gas icon mode: 0: normal; 1: blinking |
| pd.iconCoGasState | byte | CO toxic gas icon state: 0: off; 1: on. This parameter is valid only when the CO toxic gas icon mode is 0. |
| pd.iconTransSwMode | byte | Transfer switch icon mode: 0: normal; 1: blinking |
| pd.iconTransSwState | byte | Transfer switch icon state: 0: off; 1: on. This parameter is valid only when the transfer switch icon mode is 0. |
| pd.iconEcoMode | byte | ECO icon mode: 0: normal; 1: blinking |
| pd.iconEcoState | byte | ECO icon state: 0: off; 1: on. This parameter is valid only when the ECO icon mode is 0. |
| pd.iconWindGenMode | byte | Wind power generation icon mode: 0: normal; 1: blinking |
| pd.iconWindGenState | byte | Wind power generation icon |

| Field | Field's Type | Description |
| --- | --- | --- |
| | | state: 0: off; 1: on. This parameter is valid only when the wind power generation icon mode is 0. |
| pd.iconGasGenMode | byte | Oil-powered generation icon mode: 0: normal; 1: blinking |
| pd.iconGasGenState | byte | Oil-powered generation icon state: 0: off; 1: on. This parameter is valid only when the oil-powered generation icon mode is 0. |
| pd.iconSolarBracketMode | byte | Solar panel tracking bracket icon mode: 0: normal; 1: blinking |
| pd.iconSolarBracketState | byte | Solar panel tracking bracket icon state: 0: off; 1: on. This parameter is valid only when the solar panel tracking bracket icon mode is 0. |
| pd.iconSolarPanelMode | byte | Solar panel icon mode: 0: normal; 1: blinking |
| pd.iconSolarPanelState | byte | Solar panel icon state: 0: off; 1: on. This parameter is valid only when the solar panel icon mode is 0. |
| pd.lcdBrightness | int | Screen brightness<br>Range：0~100<br>Input 128(0x11111111), indicates turned on the automatic brightness adjustment |
| ems.chgState | short | Charging state |
| ems.chgCmd | short | Charge command |
| ems.dsgCmd | short | Discharge command |
| ems.chgVol | long | Charging voltage |
| ems.chgAmp | long | Charging current |
| ems.fanLevel | short | Fan level |
| ems.maxChargeSoc | short | Charge upper limit |

| Field | Field's Type | Description |
| --- | --- | --- |
| ems.bmsModel | short | BMS model |
| ems.lcdShowSoc | short | SOC on LCD |
| ems.openUpsFlag | short | UPS mode enable flag |
| ems.bmsWarningState | short | BMS warning state: bit0: hi_temp; bit1: low_temp; bit2: overload; bit3: chg_flag |
| ems.chgRemainTime | long | Remaining charging time (mins) |
| ems.dsgRemainTime | long | Remaining discharging time (mins) |
| ems.emsIsNormalFlag | short | Energy storage state: 0: sleep; 1: normal |
| ems.f32LcdShowSoc | float | SOC on LCD |
| ems.bms0Online | short | BMS online signal: BIT0: hardware online signal; BIT1: software online signal |
| ems.bms1Online | short | BMS online signal: BIT0: hardware online signal; BIT1: software online signal |
| ems.bms2Online | short | BMS online signal: BIT0: hardware online signal; BIT1: software online signal |
| ems.maxAvailableNum | short | Maximum available quantity |
| ems.openBmsIdx | short | Open BMS index |
| ems.paraVolMin | long | Minimum parallel voltage |
| ems.paraVolMax | long | Maximum parallel voltage |
| ems.minDsgSoc | short | Discharge lower limit |
| ems.minOpenOilEbSoc | short | The upper threshold of smart generator auto on<br>Range: 0~100 |
| ems.maxCloseOilEbSoc | short | The lower threshold of smart generator auto off<br>Range: 0~100 |

Example

```
// Get Delta Pro all quota values
```

```
// Request
curl -X GET https://api-e.ecoflow.com/iot-open/sign/device/quota/all?sn=DCABZ*** \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:534215ccfe93979588975630ea0ad7091c7e250ae5***'

// Response
{
    "code":"0",
    "message":"Success",
    "data":{
        "bmsMaster.soc":"100",
        "bmsMaster.temp":"34",
        "bmsMaster.inputWatts":"0",
        "bmsMaster.outputWatts":"0",
        "pd.remainTime":"14781",
        "inv.cfgAcEnabled":"0",
        "mppt.carState":"0",
        "pd.usb1Watts":"0",
        "pd.usb2Watts":"0",
        "pd.qcUsb1Watts":"0",
        ...
    }
}
```

## MQTT communication mode

### Set & Set Reply

| Usage of Topic | Topic | From | To |
| --- | --- | --- | --- |
| Set device function | /open/${certificateAccount}/${sn}/set | app | device |
| Reply to the set result | /open/${certificateAccount}/${sn}/set_reply | device | app |

Full Format Example

Set Data Format

```
{
  "id":"123456789",
  "version":"1.0",
  "operateType": "TCP",
  "timestamp": 1634841971000,
  "params": {
    "cmdSet": 32,
    "id": 33,
    "standByMode": 0
  }
}
```

Set Reply Data Format

```
{
    "id":123456789,
    "version":"1.0",
    "operateType": "TCP",
    "code": "0",
    "data": {
        "cmdSet": 32,
        "id": 33,
        "ack": 0
    }
}
```

ParamInfo Message Format

| Set Command | SetTopic's ParamInfo | SetReplyTopic's ParamInfo |
| --- | --- | --- |
| Setting the X-Boost switch | `"params":{` | `"data":{` |

| Set Command | SetTopic's ParamInfo | SetReplyTopic's ParamInfo |
| --- | --- | --- |
| | "cmdSet": 32,<br>"id": 66,<br>"enabled": 1,<br>"xboost": 0<br>} | "cmdSet": 32,<br>"id": 66,<br>"ack": 0<br>} |
| Setting the car charger switch | "params":{<br>"cmdSet": 32,<br>"id": 81,<br>"enabled": 1<br>} | "data":{<br>"cmdSet": 32,<br>"id": 81,<br>"ack": 0<br>} |
| Setting the charge level | "params":{<br>"cmdSet": 32,<br>"id": 49,<br>"maxChgSoc":<br>100<br>} | "data":{<br>"cmdSet": 32,<br>"id": 49,<br>"ack": 0<br>} |
| Setting the discharge level | "params":{<br>"cmdSet": 32,<br>"id": 51,<br>"minDsgSoc": 10<br>} | "data":{<br>"cmdSet": 32,<br>"id": 51,<br>"ack": 0<br>} |
| Setting the car input current | "params":{<br>"cmdSet": 32,<br>"id": 71,<br>"currMa": 0<br>} | "data":{<br>"cmdSet": 32,<br>"id": 71,<br>"ack": 0<br>} |
| Setting the beep switch | "params":{<br>"cmdSet": 32,<br>"id": 38,<br>"enabled": 1<br>} | "data":{<br>"cmdSet": 32,<br>"id": 38,<br>"ack": 0<br>} |
| Setting the screen brightness | "params":{<br>"cmdSet": 32,<br>"id": 39,<br><br>"lcdBrightness": | "data":{<br>"cmdSet": 32,<br>"id": 39,<br>"ack": 0<br>} |

| Set Command | SetTopic's ParamInfo | SetReplyTopic's ParamInfo |
| --- | --- | --- |
| | 100<br>} | |
| Setting the lower upper threshold percentage of smart generator auto on | "params":{<br>"cmdSet": 32,<br>"id": 52,<br>"openOilSoc":<br>52<br>} | "data":{<br>"cmdSet": 32,<br>"id": 52,<br>"ack": 0<br>} |
| Setting the upper threshold percentage of smart generator auto off | "params":{<br>"cmdSet": 32,<br>"id": 53,<br>"closeOilSoc":<br>10<br>} | "data":{<br>"cmdSet": 32,<br>"id": 53,<br>"ack": 0<br>} |
| Setting the unit timeout | "params":{<br>"cmdSet": 32,<br>"id": 33,<br>"standByMode":<br>0<br>} | "data":{<br>"cmdSet": 32,<br>"id": 33,<br>"ack": 0<br>} |
| Setting the screen timeout | "params":{<br>"cmdSet": 32,<br>"id": 39,<br>"lcdTime": 60<br>} | "data":{<br>"cmdSet": 32,<br>"id": 39,<br>"lcdTime": 60<br>} |
| Setting the AC standby time | "params":{<br>"cmdSet": 32,<br>"id": 153,<br>"standByMins":<br>720<br>} | "data":{<br>"cmdSet": 32,<br>"id": 153,<br>"ack": 0<br>} |
| AC charging settings | "params":{<br>"cmdSet": 32,<br>"id": 69,<br>"slowChgPower":<br>0 | "data":{<br>"cmdSet": 32,<br>"id": 69,<br>"ack": 0<br>} |

| Set Command | SetTopic's ParamInfo | SetReplyTopic's ParamInfo |
| --- | --- | --- |
| | }<br> | |
| PV charging type settings | "params":{<br>"cmdSet": 32,<br>"id": 82,<br>"chgType": 0<br>} | "data":{<br>"cmdSet": 32,<br>"id": 82,<br>"ack": 0<br>} |
| Bypass AC auto start settings | "params":{<br>"cmdSet": 32,<br>"id": 84,<br>"enabled": 0<br>} | "data":{<br>"cmdSet": 32,<br>"id": 84,<br>"ack": 0<br>} |

**Get & Get Reply**

Not Support

**Report Device Quota**

| Usage of Topic | Topic | From | To |
| --- | --- | --- | --- |
| Report device quota | /open/${certificateAccount}/${sn}/quota | device | app |

Please see HTTP GetAllQuotaResponse for the fields definition.

Example

```
{
    "id" :123456789,
    "version" : "1.0" ,
    "timestamp" :111111111,
    "params" :{
        "pd.iconWifiMode" : "0" ,
        "mppt.faultCode" : "0" ,
        "ems.minDsgSoc" : "0" ,
        "pd.iconOverloadState" : "0" ,
        "ems.emsIsNormalFlag" : "1" ,
        ...
    }
}
```


## Was this page helpful?

☺ YES ☹ NO

## Was this page helpful?

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